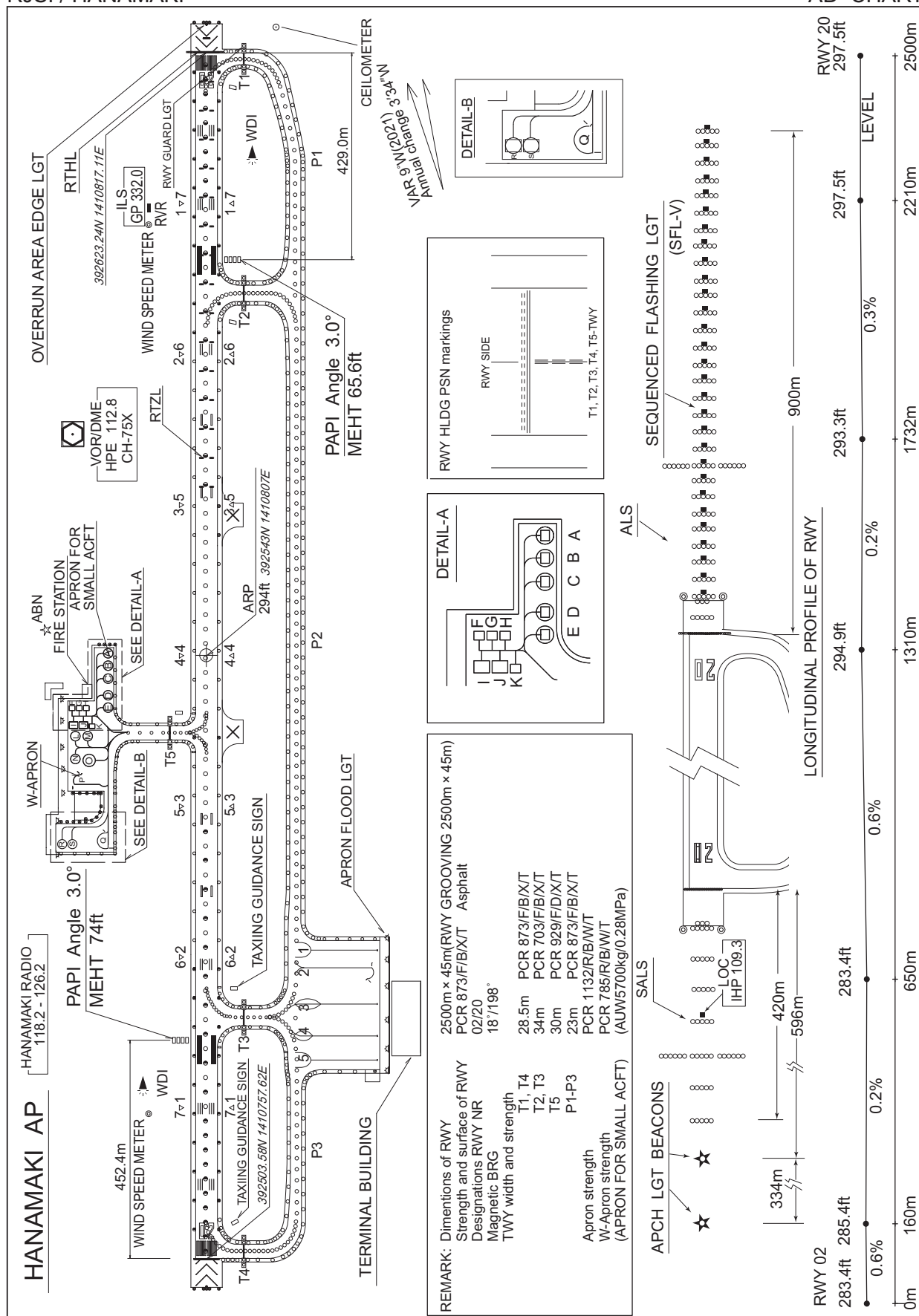


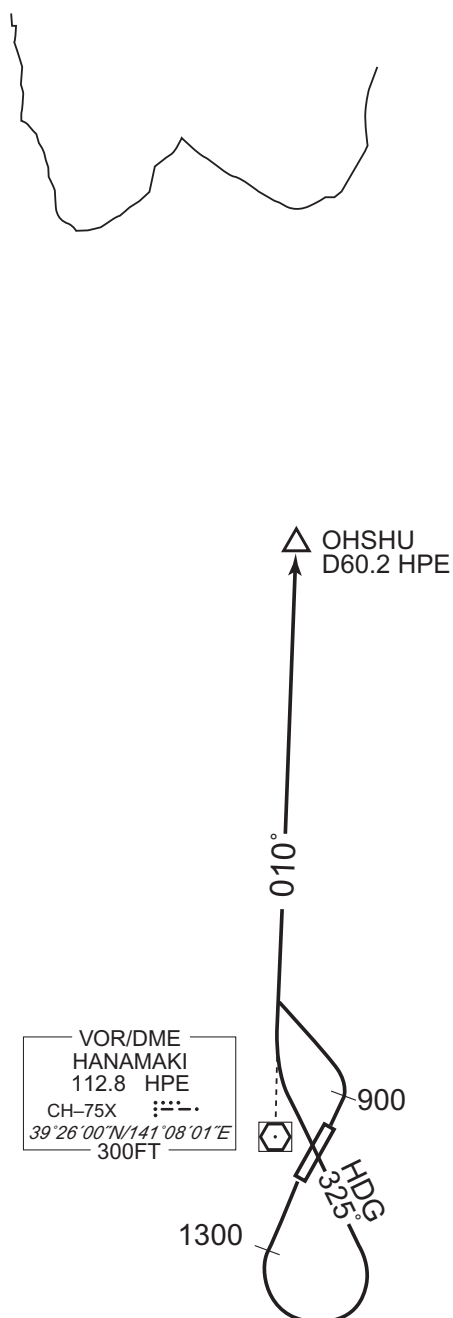
AD CHART



RJSI / HANAMAKI

SID

...to intercept and proceed via HPE R010 to OHSHU.



CHANGE : PROC course. PROC renamed.

STANDARD DEPARTURE CHART -INSTRUMENT

RJSI / HANAMAKI

SID

NIIGATA SIX DEPARTURE

RWY 02 : Climb RWY HDG to HPE 3.5 DME, turn right...

RWY 20 : Climb RWY HDG to HPE 3.5 DME, turn left...

...proceed to HPE VOR/DME, via HPE R236 to HPE 50.0DME(GTC 79.6DME),
via GTC R055 to GTC.Cross HPE VOR/DME at or above 2200 FT, cross HPE R236/50.0DME
(GTC R055/79.6DME) at or above 12000 FT.

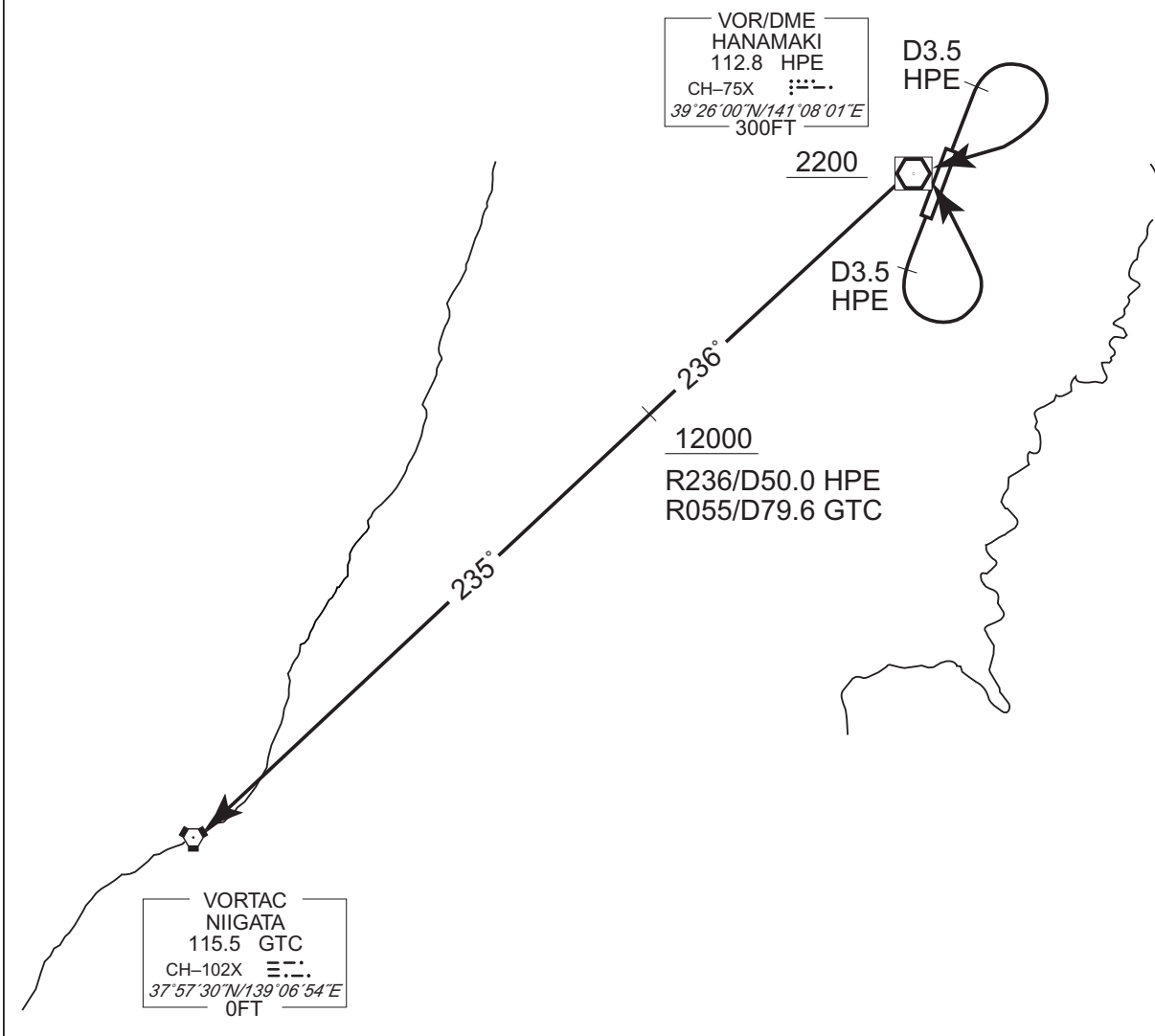
Note RWY02 : 4.5% climb gradient required up to 2400FT.

OBST ALT 1641FT located at 4.1NM 091° FM end of RWY02.

RWY20 : 3.9% climb gradient required up to 1100FT.

OBST ALT 722FT located at 2.8NM 166° FM end of RWY20.

CHANGE : SID renamed. ALT restriction at R236/D50.0 HPE, R055/D79.6 GTC.



STANDARD DEPARTURE CHART -INSTRUMENT

RJSI / HANAMAKI

SID

HANAMAKI REVERSAL THREE DEPARTURE

RWY 02 : Climb RWY HDG to 700FT, via HPE R022 to 6.0 DME, turn right...

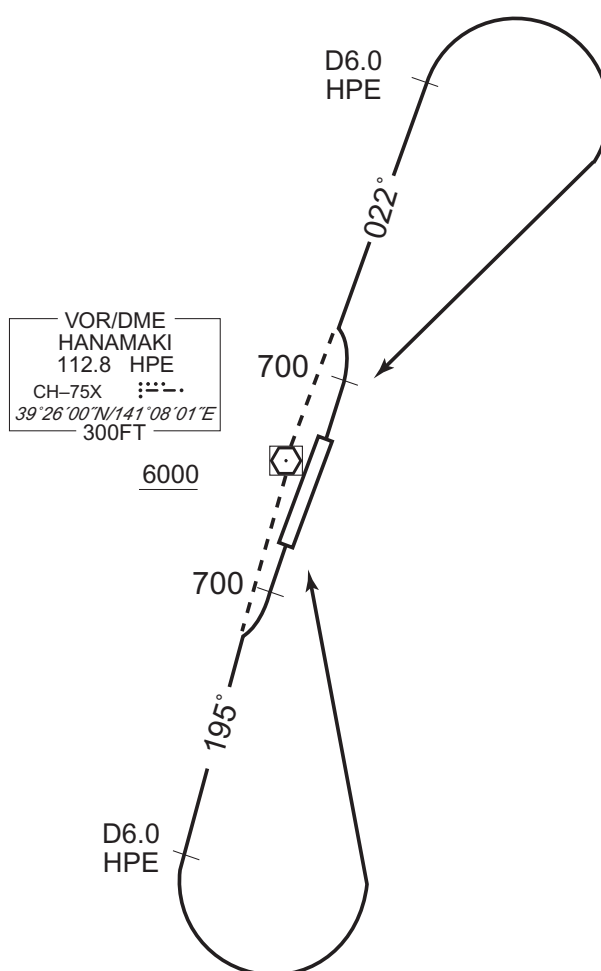
RWY 20 : Climb RWY HDG to 700FT, via HPE R195 to 6.0 DME, turn left...

...direct to HPE VOR/DME.

Cross HPE VOR/DME at or above 6000FT.

Note RWY02 : 5.0% climb gradient required up to 3200FT.

OBST ALT 2691FT located at 9.1NM 058° FM end of RWY02.



CHANGE : PROC renamed. PROC course.

STANDARD DEPARTURE CHART -INSTRUMENT

RJSI / HANAMAKI

RNAV SID

SAMBO ONE DEPARTURE

RNP1

Note GNSS required.

VAR 9°W

RWY02 : Climb on HDG 019° at or above 700FT, direct to AKARI, to DASCO to SAMBO.

RWY20 : Climb on HDG 199° at or above 1300FT, turn right direct to SI010, to AKARI, to DASCO to SAMBO .

Note RWY02 : 4.0% climb gradient required up to 700FT.

OBST ALT 318FT located at 0.2NM 061° FM end of RWY02.

RWY20 : 4.0% climb gradient required up to 2700FT.

OBST ALT 3117FT located at 10.7NM 351° FM end of RWY20.

RWY02

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course M(T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	019 (010.7)	-8.7	-	-	+700	-	-	RNP1
002	DF	AKARI	-	-	-8.7	-	-	-	-	-	RNP1
003	TF	DASCO	-	009 (000.8)	-8.7	7.9	-	-	-	-	RNP1
004	TF	SAMBO	-	345 (336.6)	-8.7	21.9	-	-	-	-	RNP1

RWY20

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course M(T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	199 (190.7)	-8.7	-	-	+1300	-	-	RNP1
002	DF	SI010	-	-	-8.7	-	R	-	-	-	RNP1
003	TF	AKARI	-	025 (016.7)	-8.7	17.0	-	-	-	-	RNP1
004	TF	DASCO	-	009 (000.8)	-8.7	7.9	-	-	-	-	RNP1
005	TF	SAMBO	-	345 (336.6)	-8.7	21.9	-	-	-	-	RNP1

CHANGE : Description of VOR/DME.

STANDARD DEPARTURE CHART -INSTRUMENT

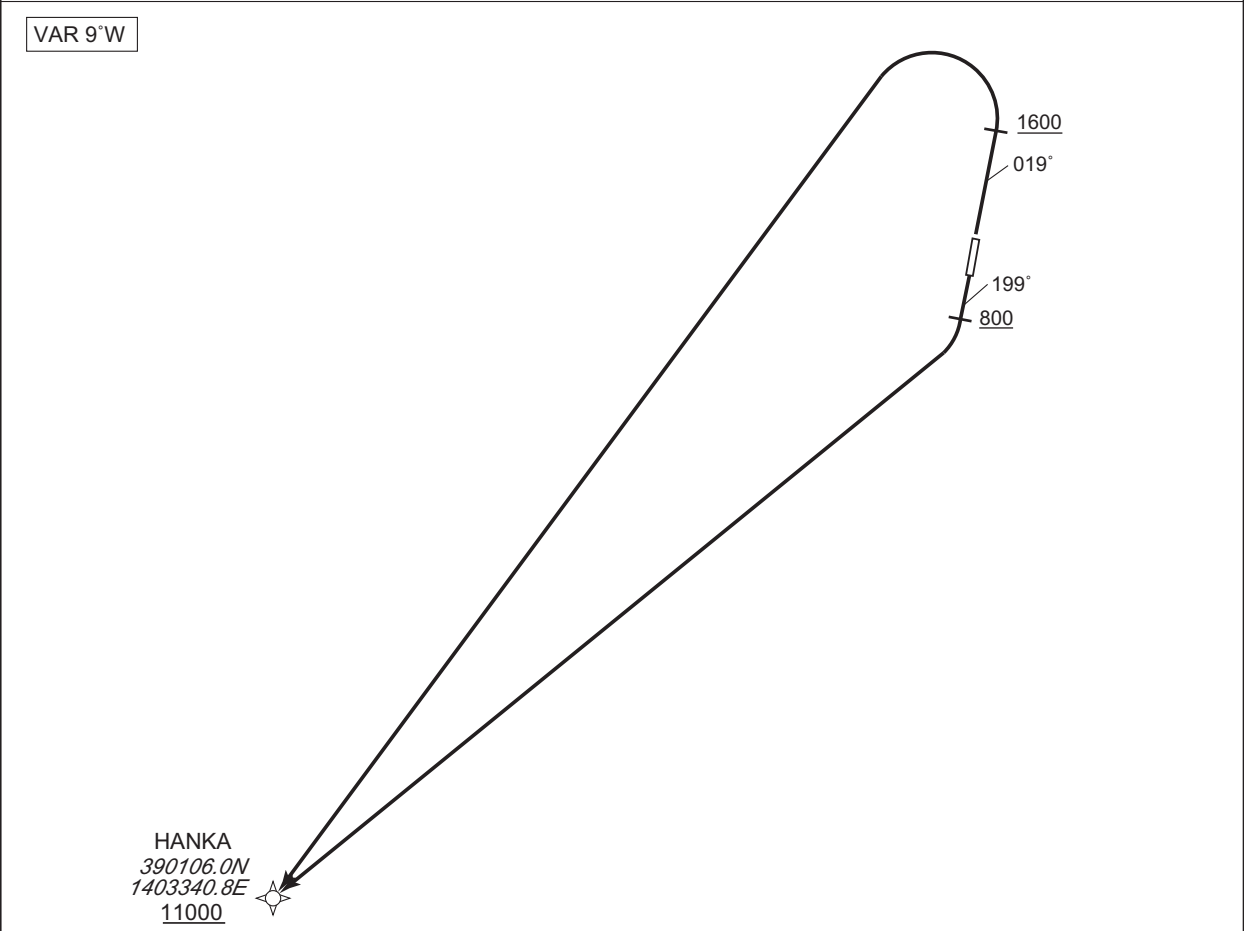
RJSI / HANAMAKI

RNAV SID

HANKA ONE DEPARTURE

RNP1

Note GNSS required



RWY02 : Climb on HDG 019° at or above 1600FT, turn left direct to HANKA, at or above 11000FT.
RWY20 : Climb on HDG 199° at or above 800FT, turn right direct to HANKA, at or above 11000FT.
Note RWY02: 5.0% climb gradient required up to 3600FT.
OBST ALT 1936FT located at 5.5NM 340° FM end of RWY02.
OBST ALT 3018FT located at 8.2NM 310° FM end of RWY02.
RWY20: 5.0% climb gradient required up to 5400FT.
OBST ALT 4593FT located at 18.2NM 227° FM end of RWY20.
OBST ALT 5151FT located at 20.8NM 232° FM end of RWY20.

CHANGE : Description of VOR/DME:

RWY02											
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	—	—	019 (010.7)	-8.7	—	—	+1600	—	—	RNP1
002	DF	HANKA	—	—	-8.7	—	L	+11000	—	—	RNP1

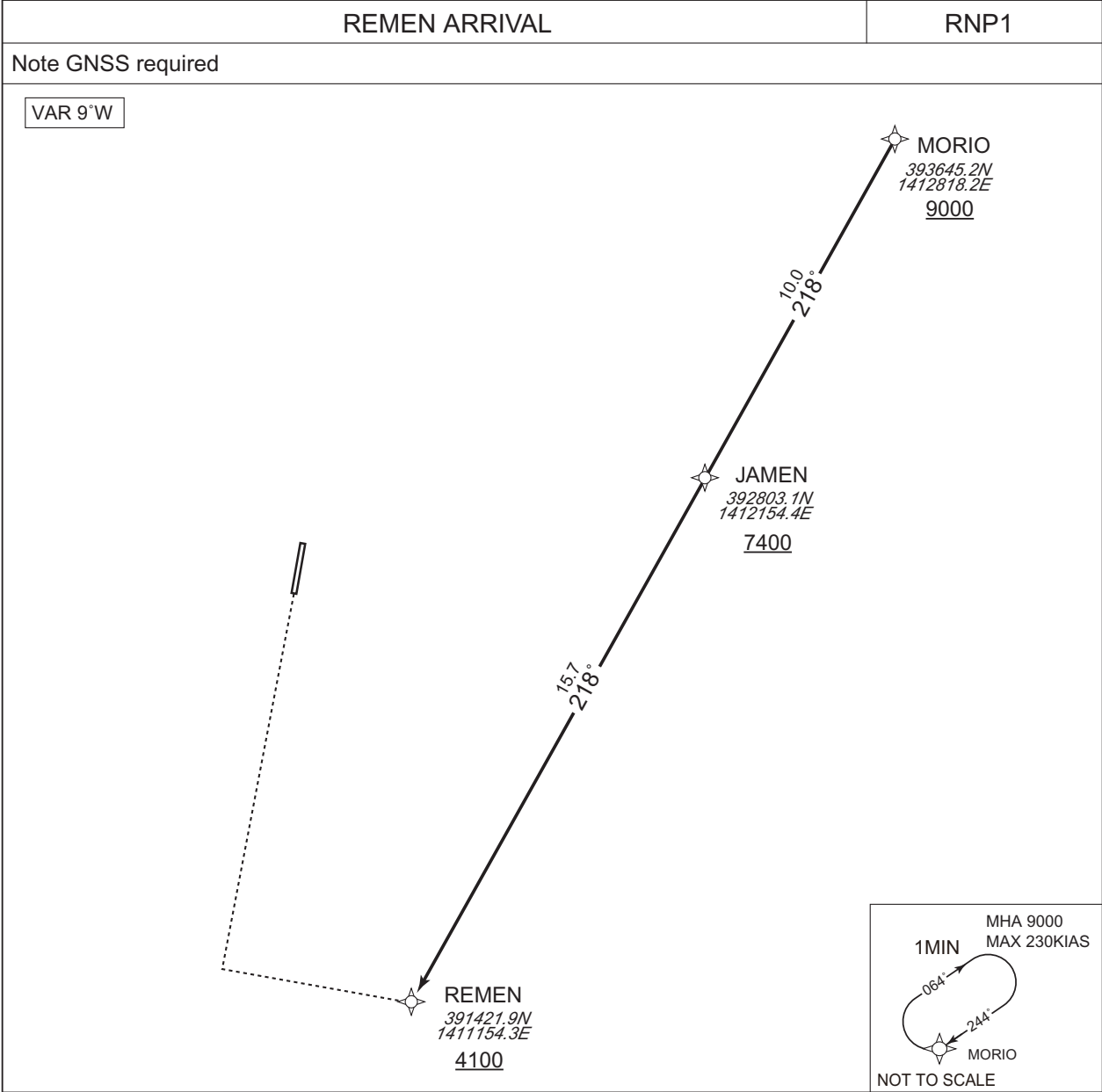
RWY20											
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	—	—	199 (190.7)	-8.7	—	—	+800	—	—	RNP1
002	DF	HANKA	—	—	-8.7	—	R	+11000	—	—	RNP1

INTENTIONALLY LEFT BLANK

STANDARD ARRIVAL CHART - INSTRUMENT

RJSI / HANAMAKI

RNAV STAR RWY02



From MORIO at or above 9000FT, to JAMEN at or above 7400FT, to REMEN at or above 4100FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	MORIO	—	—	-8.8	—	—	+9000	—	—	RNP1
002	TF	JAMEN	—	218 (209.6)	-8.8	10.0	—	+7400	—	—	RNP1
003	TF	REMEN	—	218 (209.5)	-8.8	15.7	—	+4100	—	—	RNP1

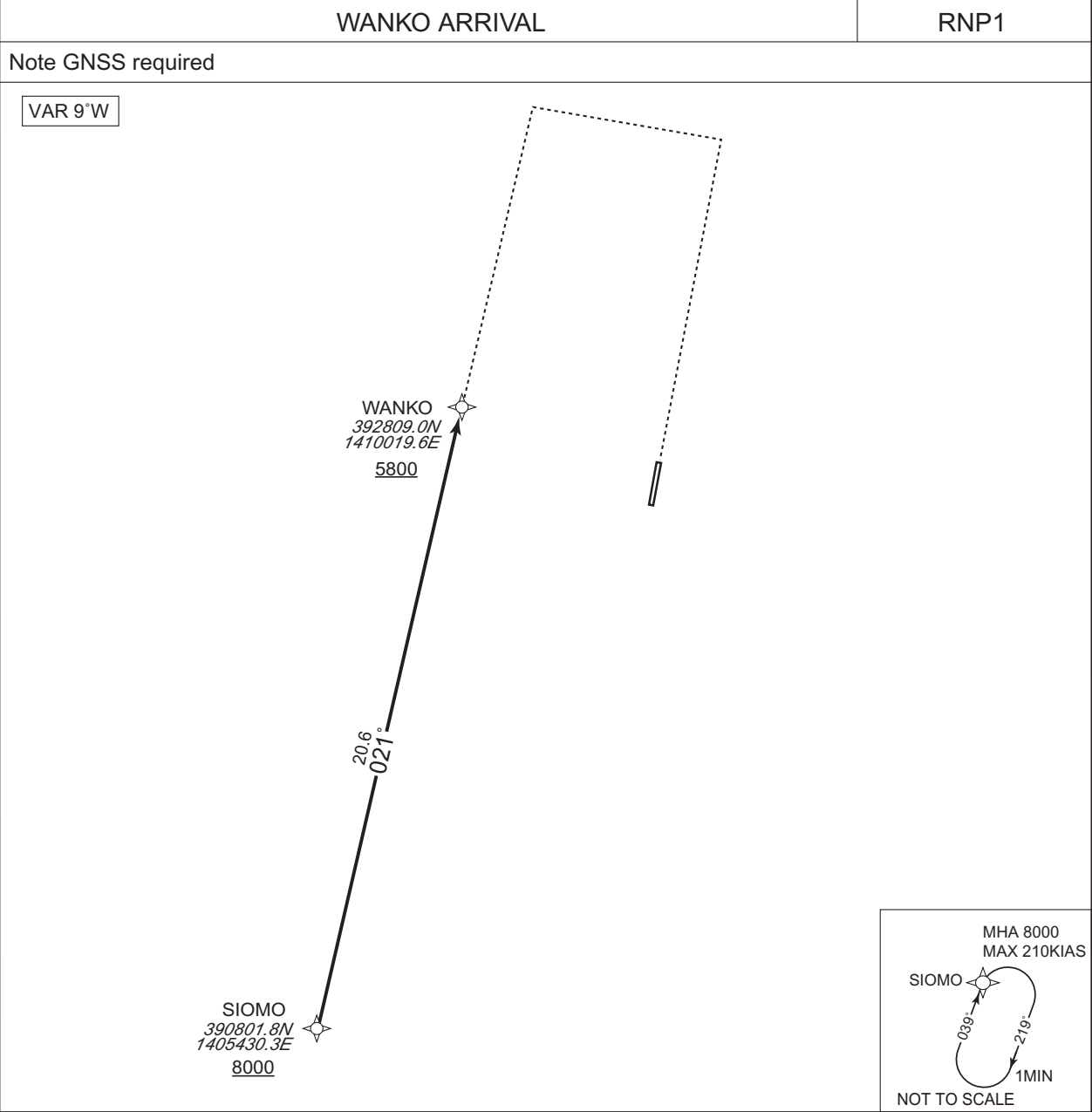
Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	MORIO	244 (235.7)	-8.8	1.0(-14000)	R	9000	FL140	-230(-14000)	RNP1

CHANGE : Description of VOR/DME.

STANDARD ARRIVAL CHART - INSTRUMENT

RJSI / HANAMAKI

RNAV STAR RWY20



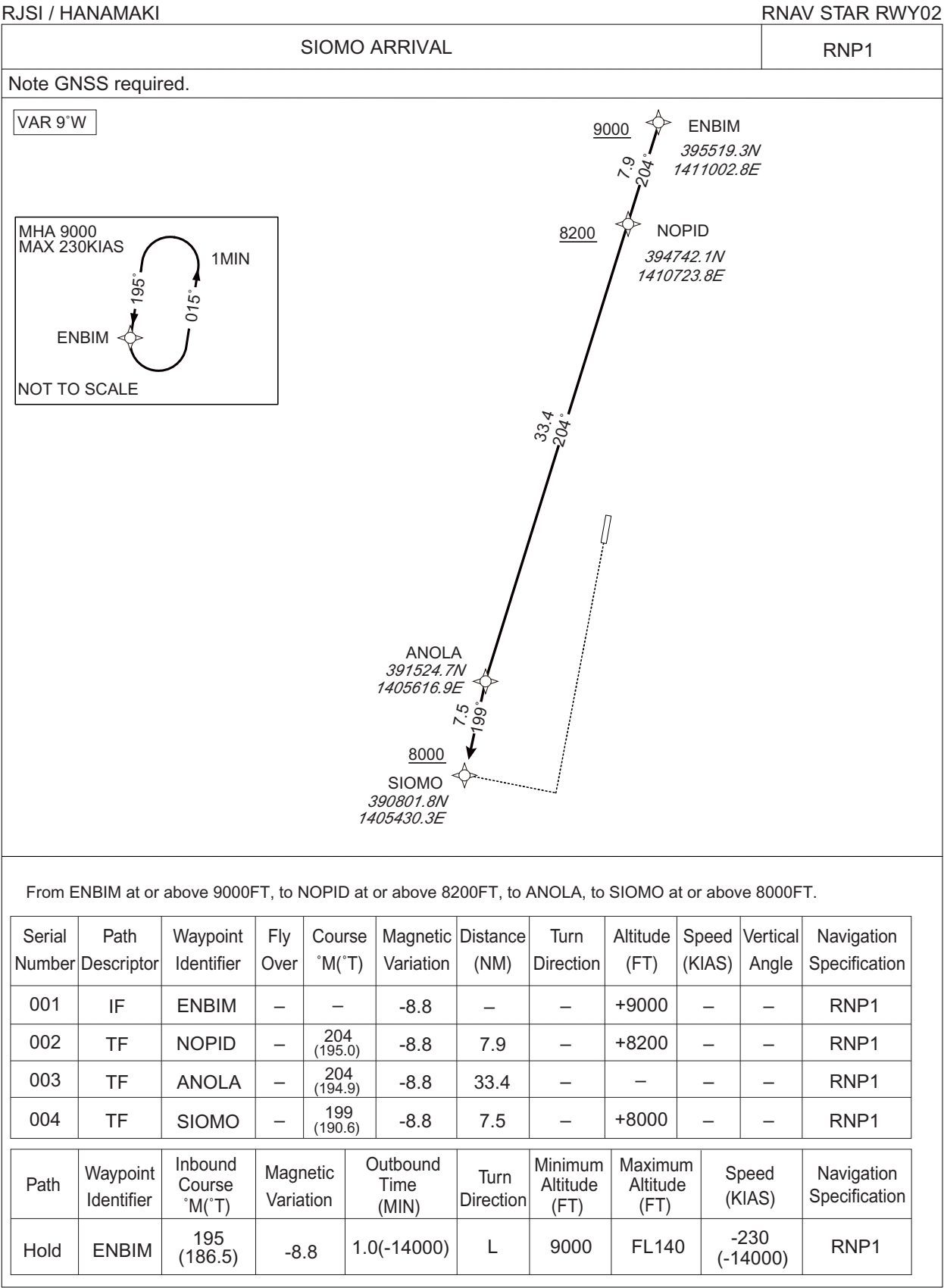
From SIOMO at or above 8000FT, to WANKO at or above 5800FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	SIOMO	—	—	-8.8	—	—	+8000	—	—	RNP1
002	TF	WANKO	—	021 (012.6)	-8.8	20.6	—	+5800	—	—	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	SIOMO	039 (030.1)	-8.8	1.0(-14000)	R	8000	FL140	-210(-14000)	RNP1

CHANGE : Description of VOR/DME.

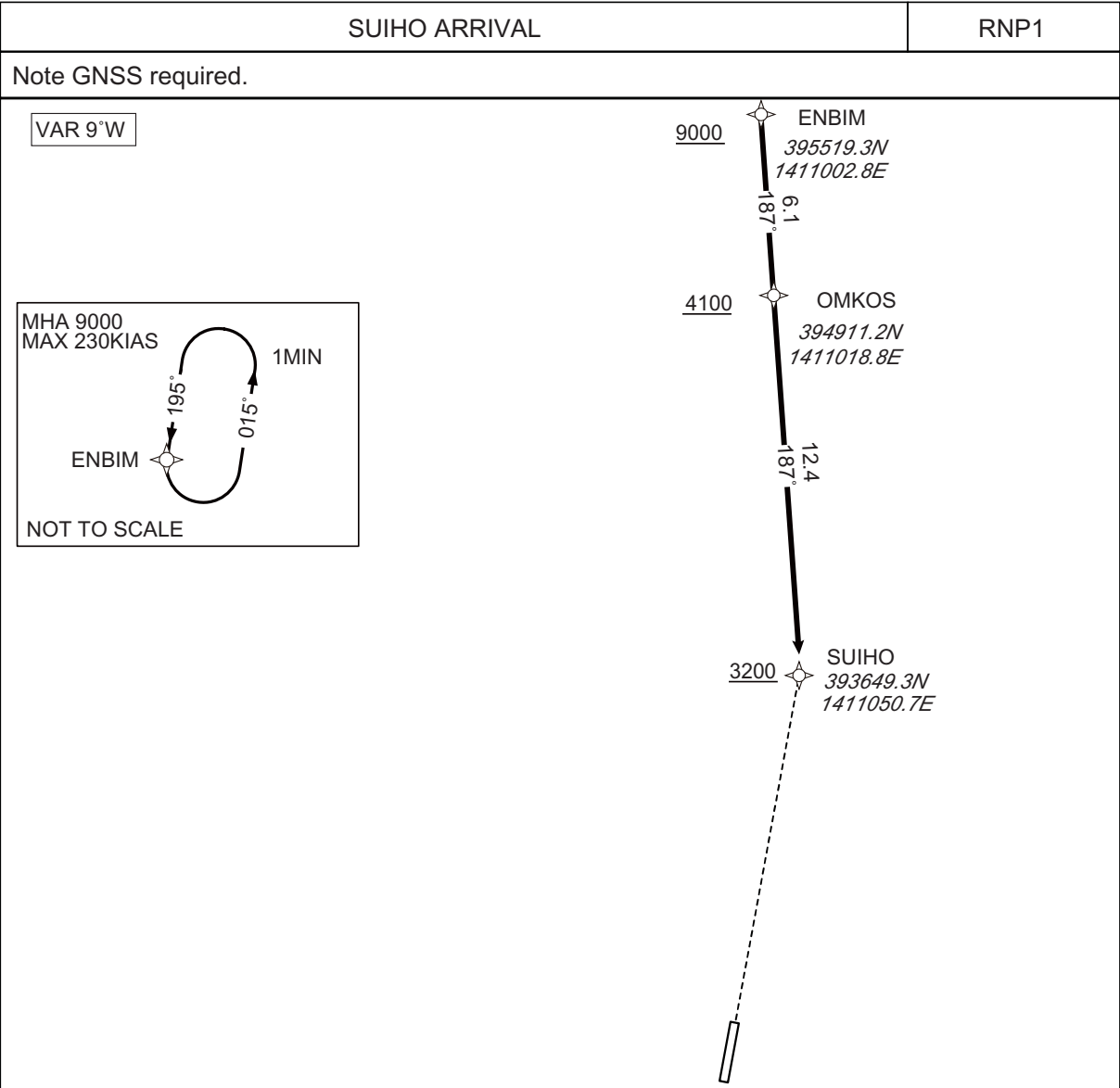
STANDARD ARRIVAL CHART - INSTRUMENT



STANDARD ARRIVAL CHART - INSTRUMENT

RJSI / HANAMAKI

RNAV STAR RWY20



From ENBIM at or above 9000FT, to OMKOS at or above 4100FT, to SUIHO at or above 3200FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	ENBIM	—	—	-8.8	—	—	+9000	—	—	RNP1
002	TF	OMKOS	—	187 (178.1)	-8.8	6.1	—	+4100	—	—	RNP1
003	TF	SUIHO	—	187 (178.1)	-8.8	12.4	—	+3200	—	—	RNP1

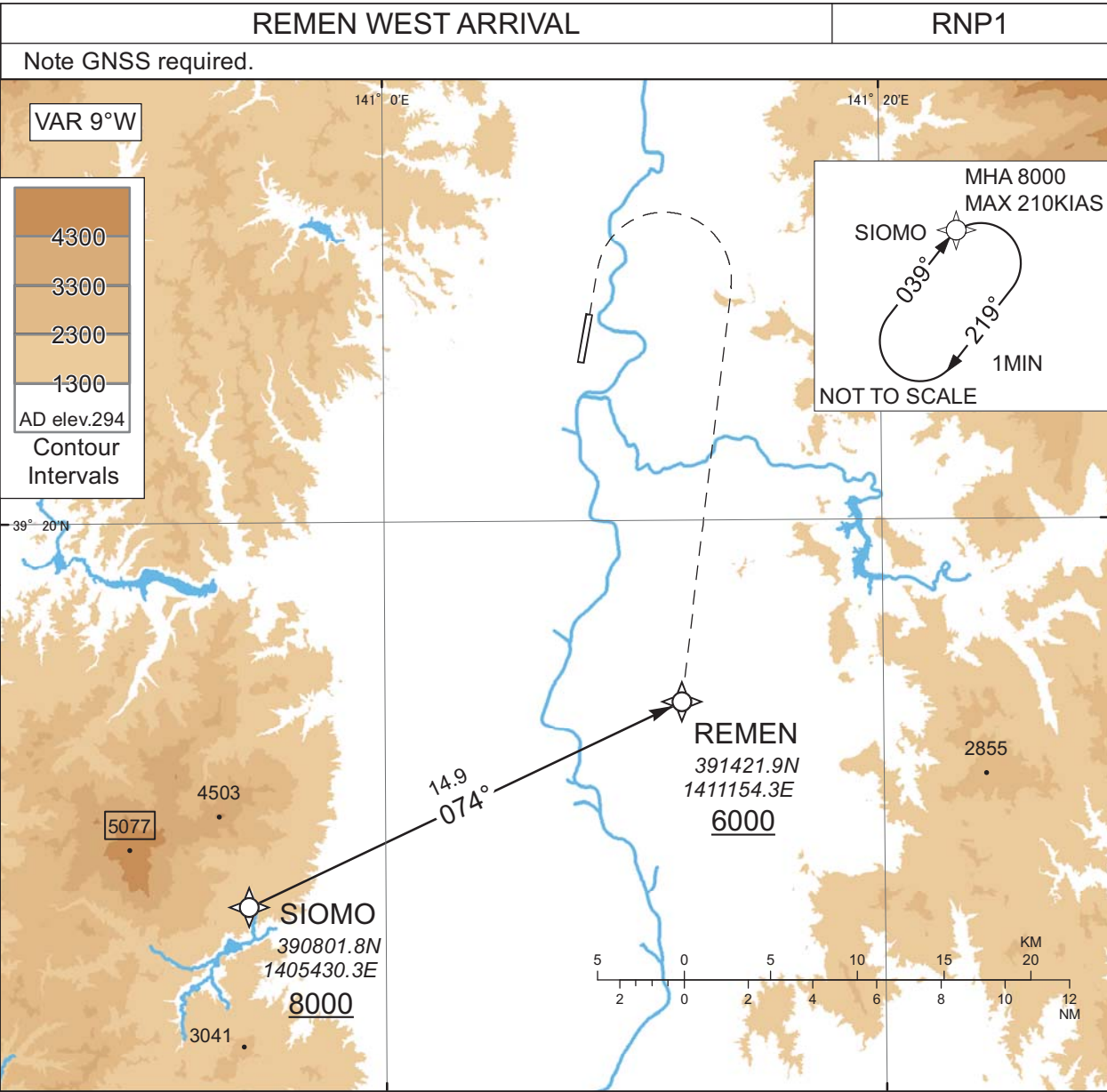
Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	ENBIM	195 (186.5)	-8.8	1.0(-14000)	L	9000	FL140	-230 (-14000)	RNP1

CHANGE : Description of VOR/DME.

STANDARD ARRIVAL CHART - INSTRUMENT

RJSI / HANAMAKI

RNAV STAR RWY20



CHANGE : Description of VOR/DME:

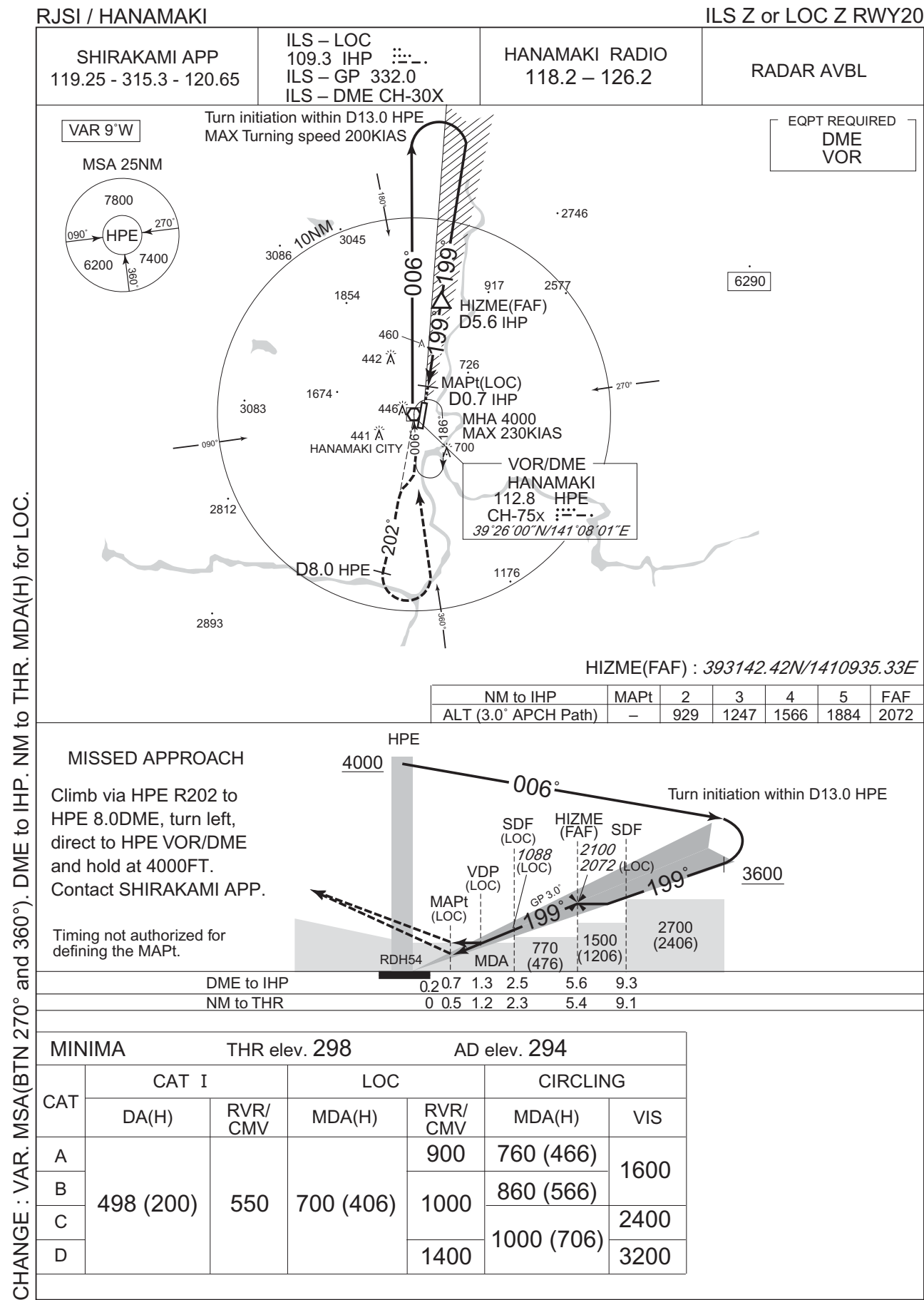
From SIOMO at or above 8000FT, to REMEN at or above 6000FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	SIOMO	-	-	-8.8	-	-	+8000	-	-	RNP1
002	TF	REMEN	-	074 (064.7)	-8.8	14.9	-	+6000	-	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	SIOMO	039 (030.1)	-8.8	1.0(-14000)	R	8000	FL140	-210(-14000)	RNP1

INTENTIONALLY LEFT BLANK

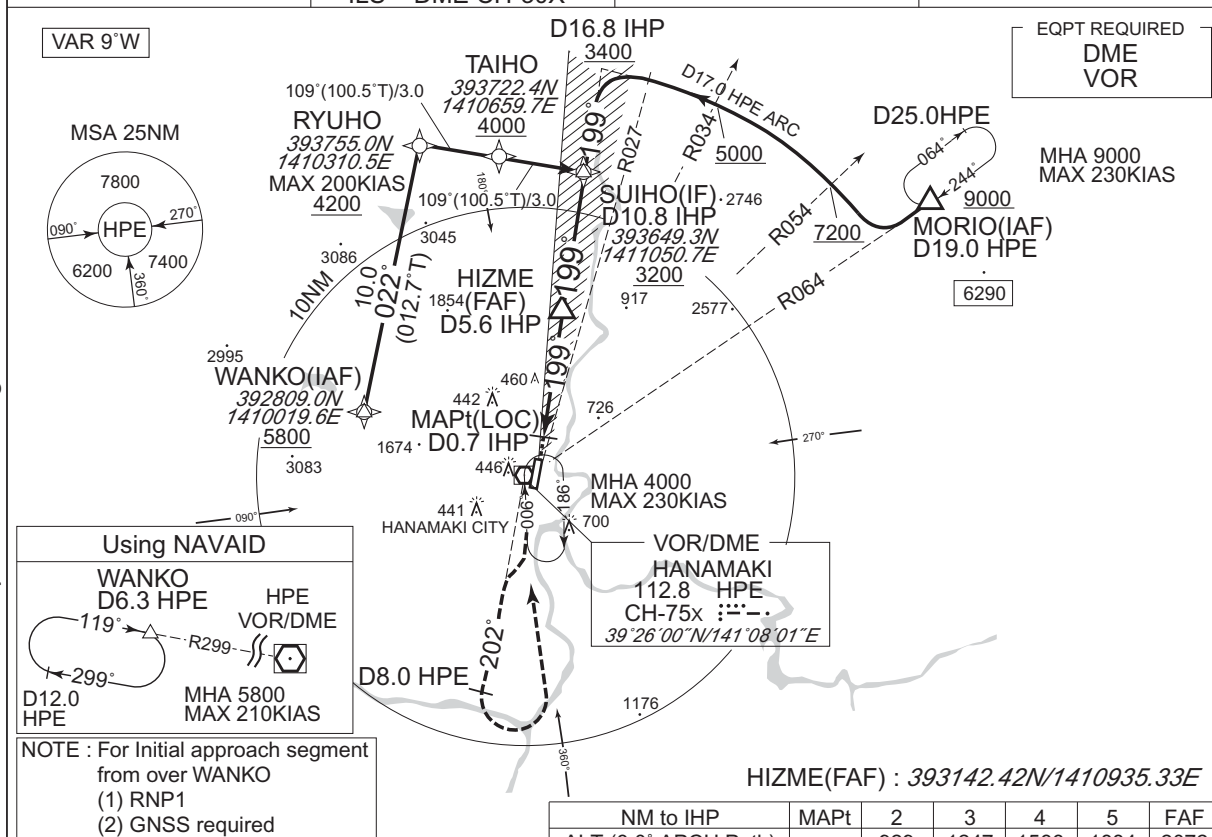
INSTRUMENT APPROACH CHART



RJSI / HANAMAKI

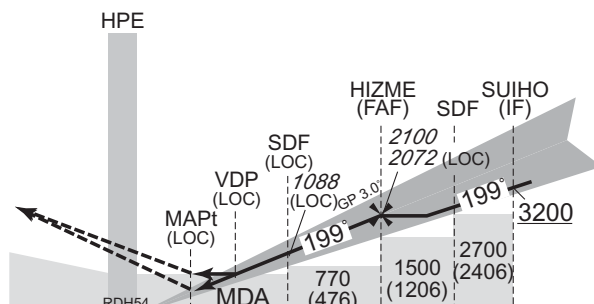
ILS Y or LOC Y RWY20

SHIRAKAMI APP 119.25 - 315.3 - 120.65	ILS – LOC 109.3 IHP ILS – GP 332.0 ILS – DME CH-30X	HANAMAKI RADIO 118.2 – 126.2	RADAR AVBL
--	--	---------------------------------	------------



Climb via HPE R202 to
HPE 8.0DME, turn left,
direct to HPE VOR/DME
and hold at 4000FT.
Contact SHIRAKAMI APP.

Timing not authorized for
defining the MAPt.

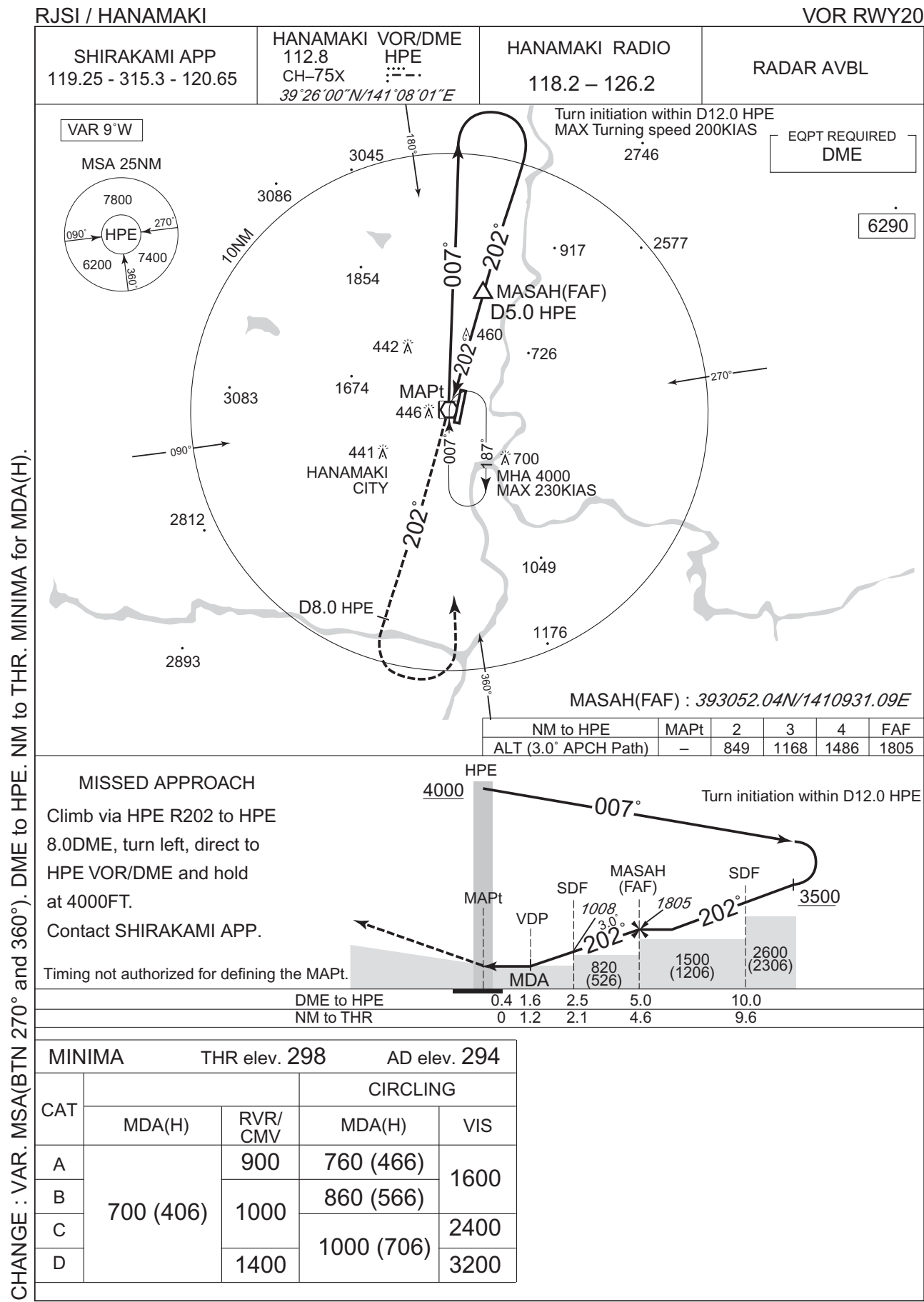


DME to IHP	0.2	0.7	1.3	2.5	5.6	9.3	10.8
NM to THR	0	0.5	1.2	2.3	5.4	9.1	10.6

MINIMA		THR elev. 298		AD elev. 294		
CAT	CAT I		LOC		CIRCLING	
	DA(H)	RVR/ CMV	MDA(H)	RVR/ CMV	MDA(H)	VIS
A	498 (200)	550	700 (406)	900	760 (466)	1600
B				1000	860 (566)	
C					1000 (706)	2400
D						1400

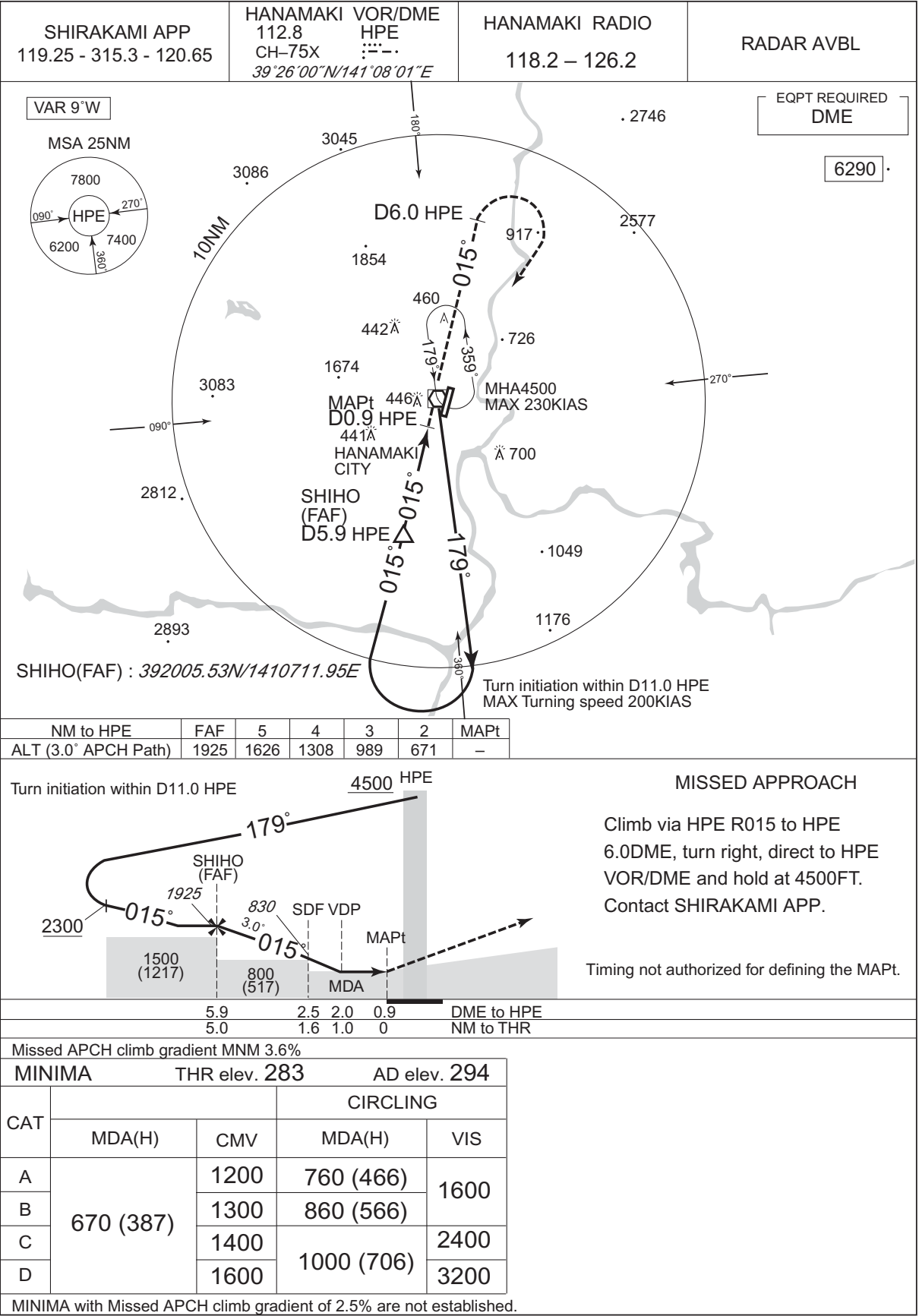
CHANGE : DME to IHP. NM to THR. MDA(H) for LOC. MSA(BTN 270° and 360°).
VAR. Course FM WANKO to RYUHO. HLDG pattern at MORIO. HLDG pattern for using NAVAID.

INSTRUMENT APPROACH CHART



INSTRUMENT APPROACH CHART

RJSI / HANAMAKI VOR RWY02

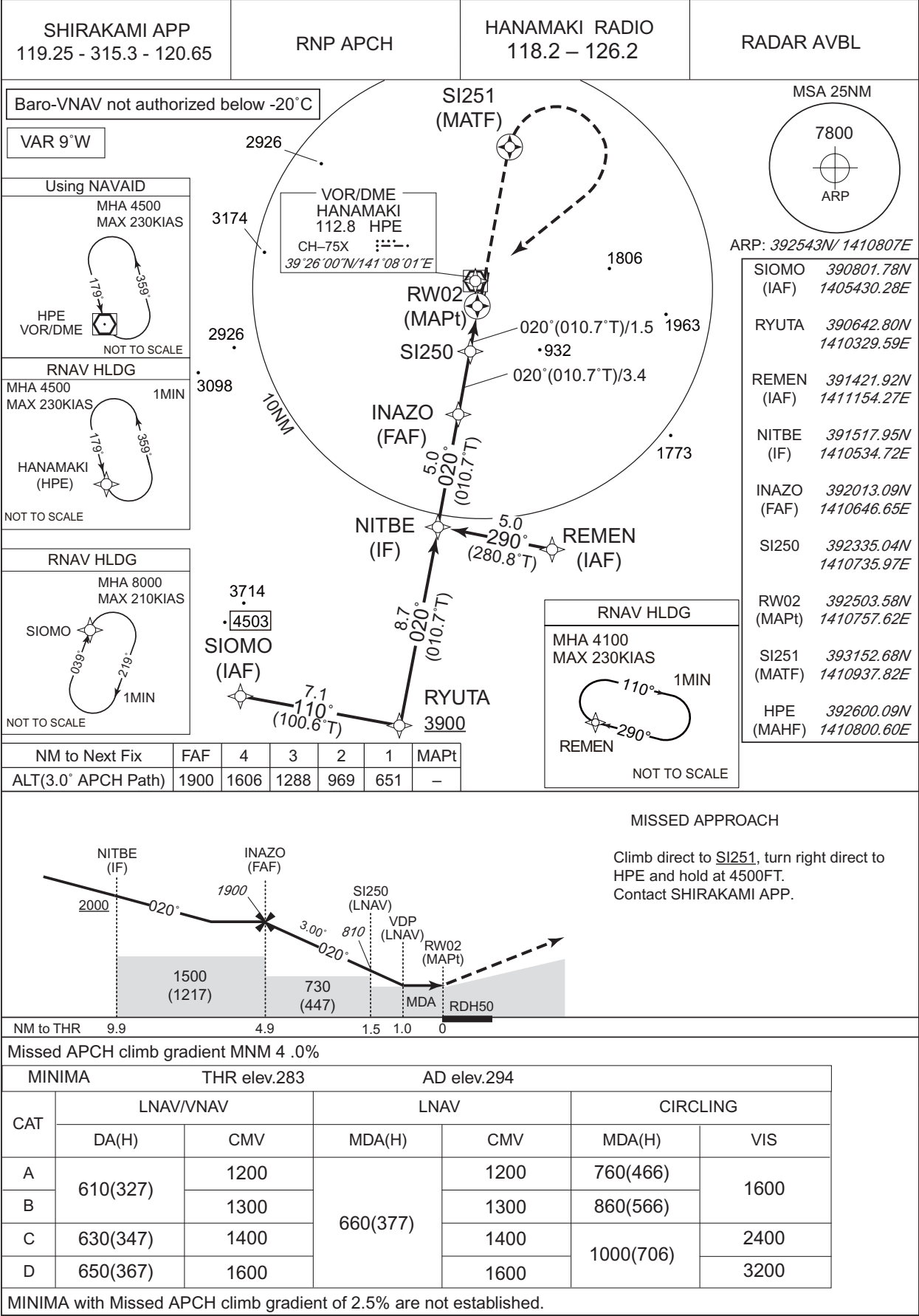


CHANGE : VAR. MSA(BTN 270° and 360°). PROC course. Missed APCH course. HLDG pattern.
OCA(H). DME to HPE. MINIMA for MDA(H).

INSTRUMENT APPROACH CHART

RJSI / HANAMAKI

RNP Z RWY02



INSTRUMENT APPROACH CHART

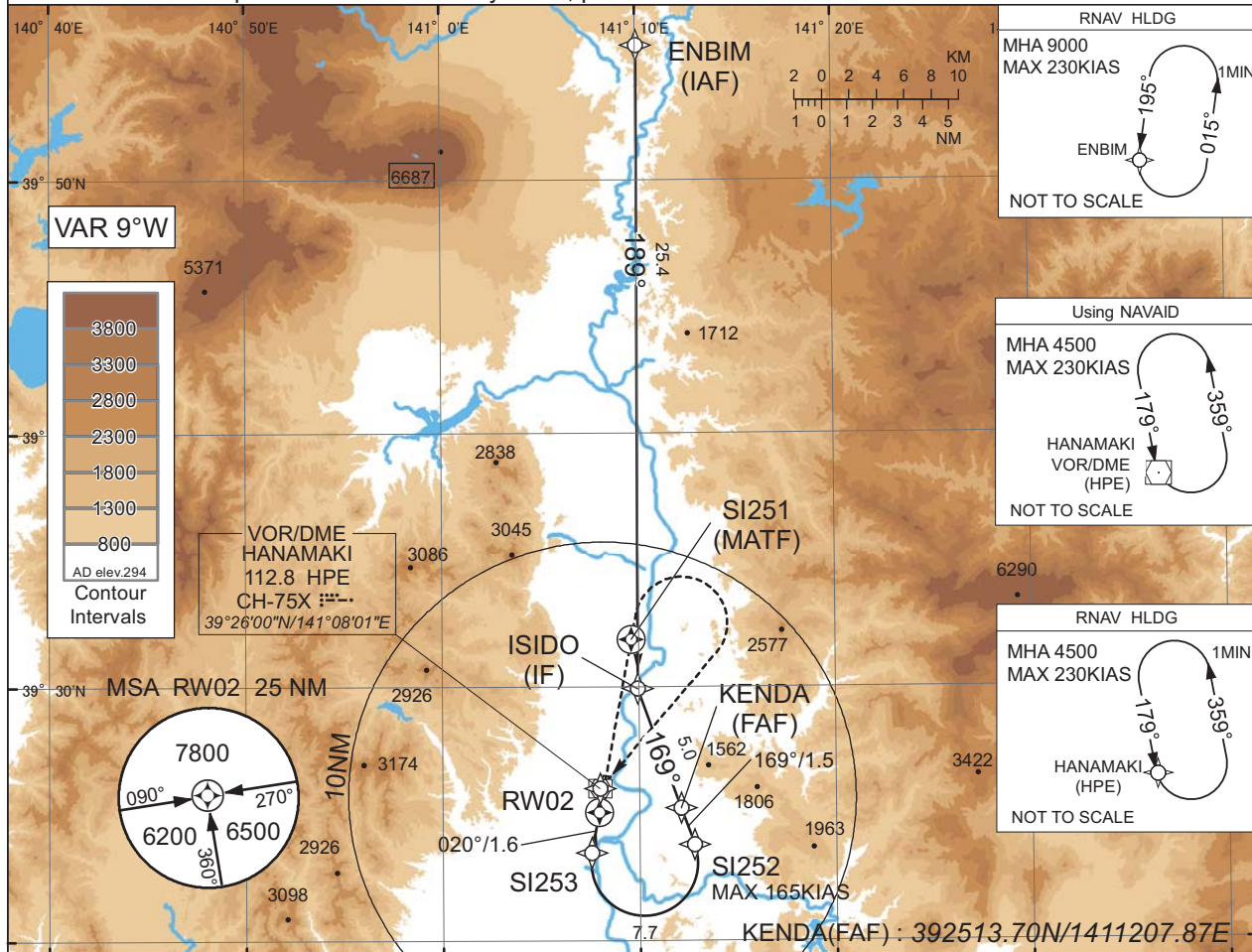
RJSI / HANAMAKI

RNP Y RWY02(AR)

SHIRAKAMI APP
119.25 - 315.3 - 120.65RNP AR
RF required.HANAMAKI RADIO
118.2 - 126.2

RADAR AVBL

For uncompensated Baro-VNAV systems, procedure not authorized below -20°C / above 45°C

ENBIM
(IAF)

9000

41.3

ISIDO
(IF)

4000

15.9

KENDA
(FAF)

3800

10.9

SI252

9.4

SI253

848

1.6

RWY02

0

MISSED APPROACH

Climb to 4500FT, direct to
SI251, turn right direct to
HPE and hold.
Contact SHIRAKAMI APP.

RDH 50

NM to THR

Missed APCH climb gradient MNM 4.0%

MINIMA	THR elev. 283	AD elev. 294
CAT	RNP 0.30	
	DA(H)	CMV
A	-	-
B	-	-
C	595(312)	1400
D		1600

MINIMA with Missed APCH climb gradient of 2.5% are not established.

Authorization Required

CHANGE : HLDG pattern for using NAVAID. RNAV HLDG at HANAMAKI.

INSTRUMENT APPROACH CHART

RJSI / HANAMAKI

RNP Y RWY02(AR)

Coding Table											
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	ENBIM	-	-	-8.8	-	-	+9000	-	-	-
002	TF	ISIDO	-	189 (180.2)	-8.8	25.4	-	+4000	-	-	1.0
003	TF	KENDA	-	169 (160.3)	-8.8	5.0	-	3800	-	-	1.0
004	TF	SI252	-	169 (160.4)	-8.8	1.5	-	3312	-165	-3.00	0.3
005	RF Center: SIRF1 r=2.10NM	SI253	-	-	-8.8	7.7	R	848	-	-3.00	0.3
006	TF	RW02	Y	020 (010.7)	-8.8	1.6	-	333	-	-3.00/50	0.3
007	DF	SI251	Y	-	-8.8	-	-	-	-	-	1.0
008	DF	HPE	-	-	-8.8	-	R	4500	-	-	1.0

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	ENBIM	195 (186.5)	-8.8	1.0 (-14000)	L	9000	FL140	-230 (-14000)	1.0
Hold	HPE	179 (169.7)	-8.9	1.0 (-14000)	L	4500	FL140	-230 (-14000)	1.0

Waypoint Coordinates				
Waypoint Identifier		Coordinates	RF Arc Center Identifier	Coordinates
ENBIM		395519.34N / 1411002.82E	SIRF1	392304.37N / 1411014.42E
ISIDO		392956.45N / 1410957.05E		
KENDA		392513.70N / 1411207.87E		
SI252		392347.07N / 1411247.89E		
SI253		392327.97N / 1410734.24E		
RW02		392503.58N / 1410757.62E		
SI251		393152.68N / 1410937.82E		
HPE		392600.09N / 1410800.60E		

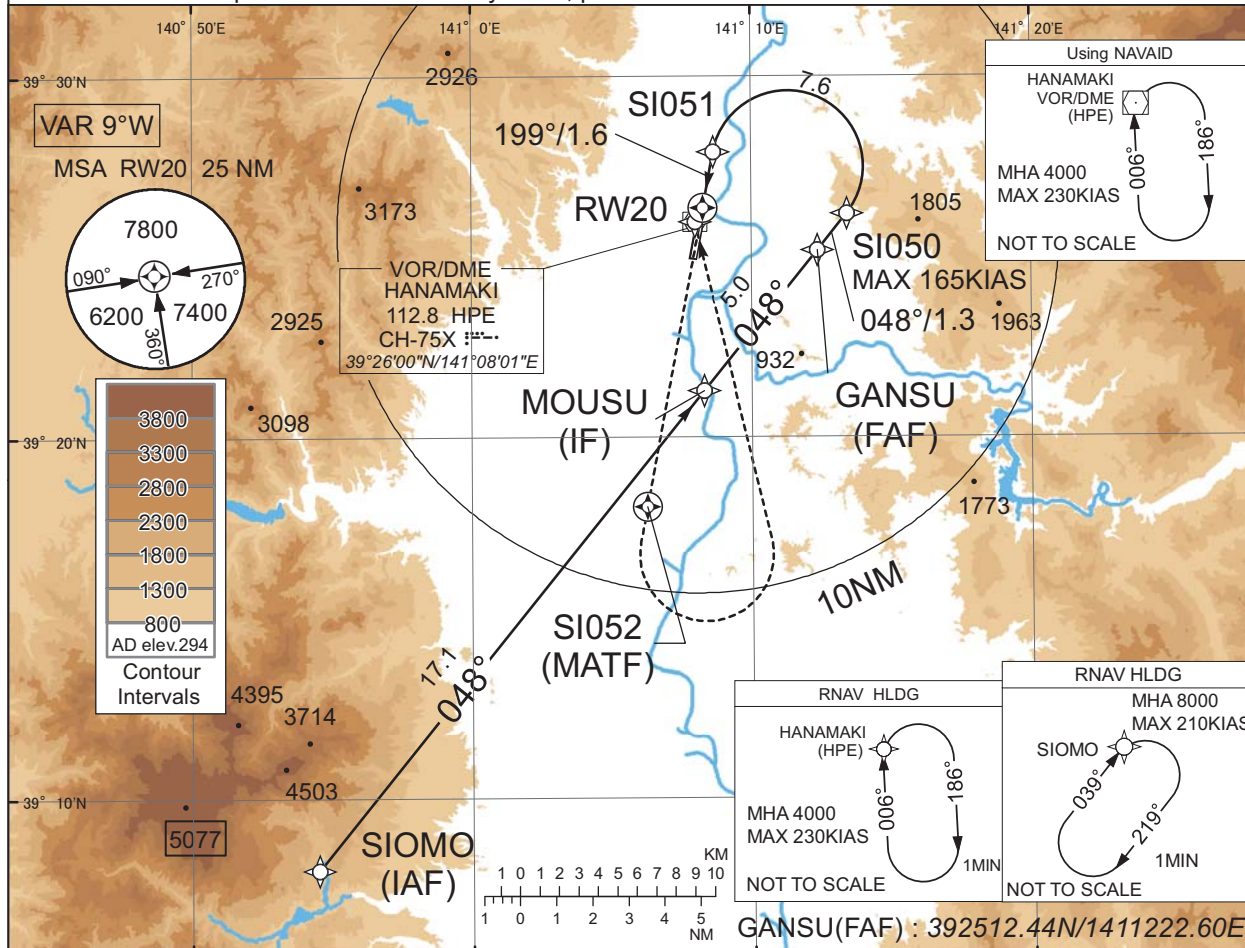
CHANGE : HLDG at HPE. VAR.

RJSI / HANAMAKI

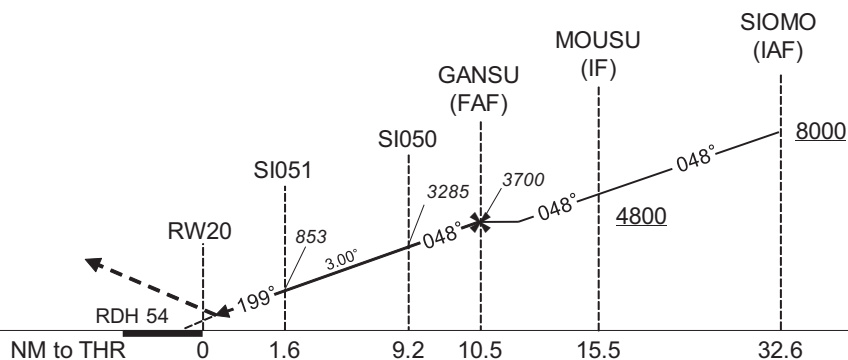
RNP Z RWY20(AR)

SHIRAKAMI APP 119.25 - 315.3 - 120.65	RNP AR RF required.	HANAMAKI RADIO 118.2 - 126.2	RADAR AVBL
--	------------------------	---------------------------------	------------

For uncompensated Baro-VNAV systems, procedure not authorized below -20°C / above 45°C



Climb to 4000FT, direct to SI052, turn left direct to HPE and hold.
Contact SHIRAKAMI APP.



MINIMA	THR elev. 298	AD elev. 294
CAT	RNP 0.30	
	DA(H)	RVR/CMV
A	-	-
B		
C	607(309)	1000
D		1400

Authorization Required

INSTRUMENT APPROACH CHART

RJSI / HANAMAKI

RNP Z RWY20(AR)

Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	SIOMO	-	-	-8.8	-	-	+8000	-	-	-
002	TF	MOUSU	-	048 (038.8)	-8.8	17.1	-	+4800	-	-	1.0
003	TF	GANSU	-	048 (038.9)	-8.8	5.0	-	3700	-	-	1.0
004	TF	SI050	-	048 (038.9)	-8.8	1.3	-	3285	-165	-3.00	0.3
005	RF Center: SIRF2 r=2.10NM	SI051	-	-	-8.8	7.6	L	853	-	-3.00	0.3
006	TF	RW20	Y	199 (190.7)	-8.8	1.6	-	352	-	-3.00/54	0.3
007	DF	SI052	Y	-	-8.8	-	-	-	-	-	1.0
008	DF	HPE	-	-	-8.8	-	L	4000	-	-	1.0

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	SIOMO	039 (030.1)	-8.8	1.0 (-14000)	R	8000	FL140	-210 (-14000)	1.0
Hold	HPE	006 (357.5)	-8.8	1.0 (-14000)	R	4000	FL140	-230 (-14000)	1.0

Waypoint Coordinates

Waypoint Identifier	Coordinates	RF Arc Center Identifier	Coordinates
SIOMO	390801.78N / 1405430.28E	SIRF2	392732.73N / 1411119.82E
MOUSU	392119.11N / 1410818.81E		
GANSU	392512.44N / 1411222.60E		
SI050	392613.23N / 1411326.21E		
SI051	392756.30N / 1410839.89E		
RW20	392623.24N / 1410817.11E		
SI052	391806.42N / 1410615.76E		
HPE	392600.09N / 1410800.60E		

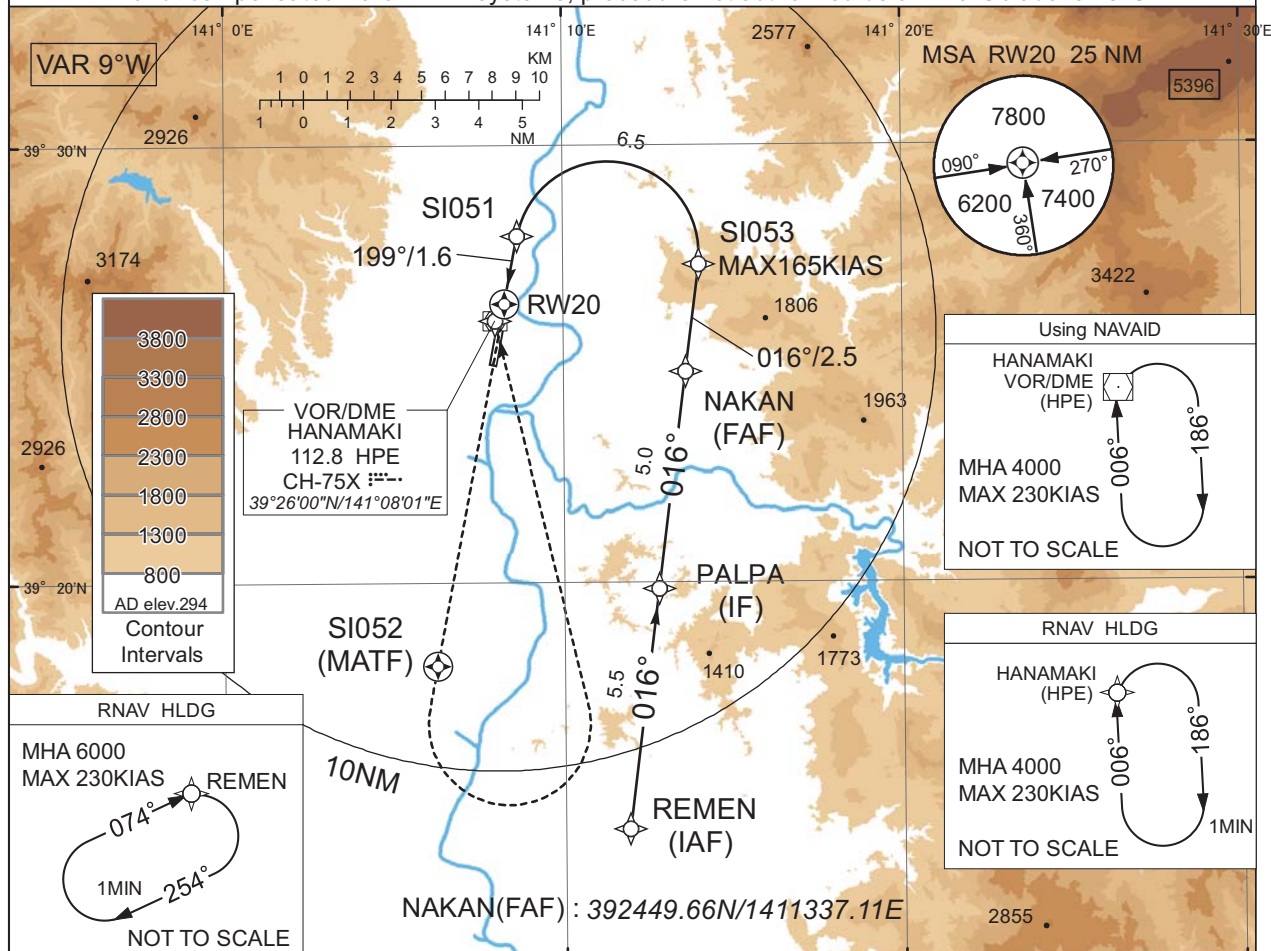
CHANGE : PROC renamed.

RJSI / HANAMAKI

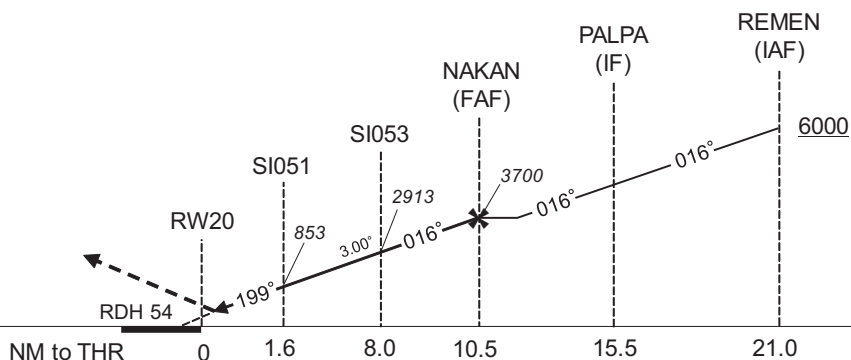
RNP Y RWY20(AR)

SHIRAKAMI APP 119.25 - 315.3 - 120.65	RNP AR RF required.	HANAMAKI RADIO 118.2 - 126.2	RADAR AVBL
--	------------------------	---------------------------------	------------

For uncompensated Baro-VNAV systems, procedure not authorized below -20°C / above 45°C



Climb to 4000FT, direct to SI052, turn left direct to HPE and hold.
Contact SHIRAKAMI APP.



MINIMA	THR elev. 298	AD elev. 294
CAT	RNP 0.30	
	DA(H)	RVR/CMV
A	-	-
B	-	-
C	607(309)	1000
D		1400

Authorization Required

INSTRUMENT APPROACH CHART

RJSI / HANAMAKI

RNP Y RWY20(AR)

Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	REMEN	-	-	-8.8	-	-	+6000	-	-	-
002	TF	PALPA	-	016 (007.2)	-8.8	5.5	-	-	-	-	1.0
003	TF	NAKAN	-	016 (007.2)	-8.8	5.0	-	3700	-	-	1.0
004	TF	SI053	-	016 (007.2)	-8.8	2.5	-	2913	-165	-3.00	0.3
005	RF Center: SIRF2 r=2.10NM	SI051	-	-	-8.8	6.5	L	853	-	-3.00	0.3
006	TF	RW20	Y	199 (190.7)	-8.8	1.6	-	352	-	-3.00/54	0.3
007	DF	SI052	Y	-	-8.8	-	-	-	-	-	1.0
008	DF	HPE	-	-	-8.8	-	L	4000	-	-	1.0

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	REMEN	074 (065.0)	-8.8	1.0 (-14000)	R	6000	FL140	-230 (-14000)	1.0
Hold	HPE	006 (357.5)	-8.8	1.0 (-14000)	R	4000	FL140	-230 (-14000)	1.0

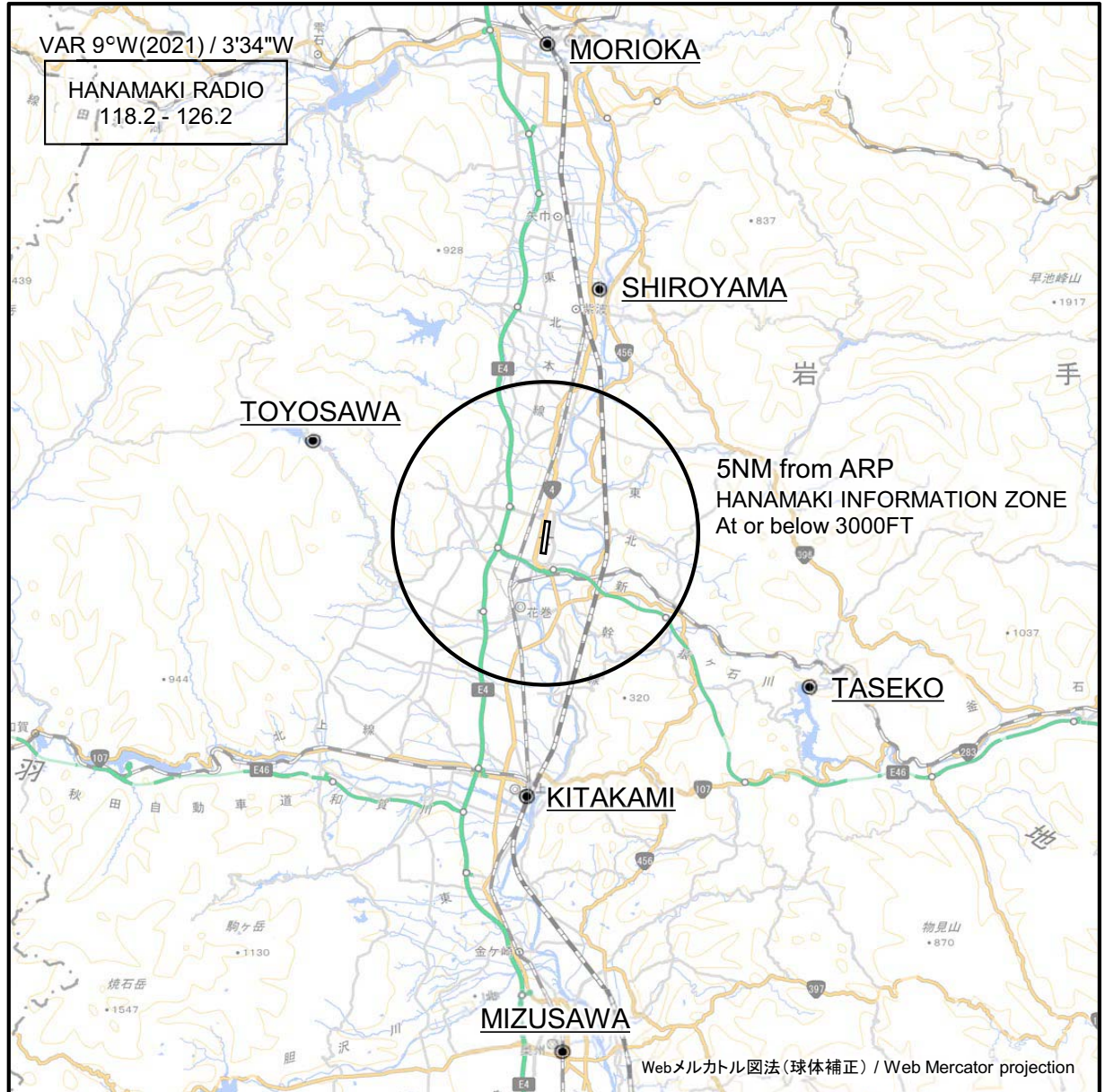
Waypoint Coordinates

Waypoint Identifier	Coordinates	RF Arc Center Identifier	Coordinates
REMEN	391421.92N / 1411154.27E	SIRF2	392732.73N / 1411119.82E
PALPA	391951.68N / 1411248.23E		
NAKAN	392449.66N / 1411337.11E		
SI053	392716.81N / 1411401.29E		
SI051	392756.30N / 1410839.89E		
RW20	392623.24N / 1410817.11E		
SI052	391806.42N / 1410615.76E		
HPE	392600.09N / 1410800.60E		

CHANGE : PROC renamed.

RJSI / HANAMAKI

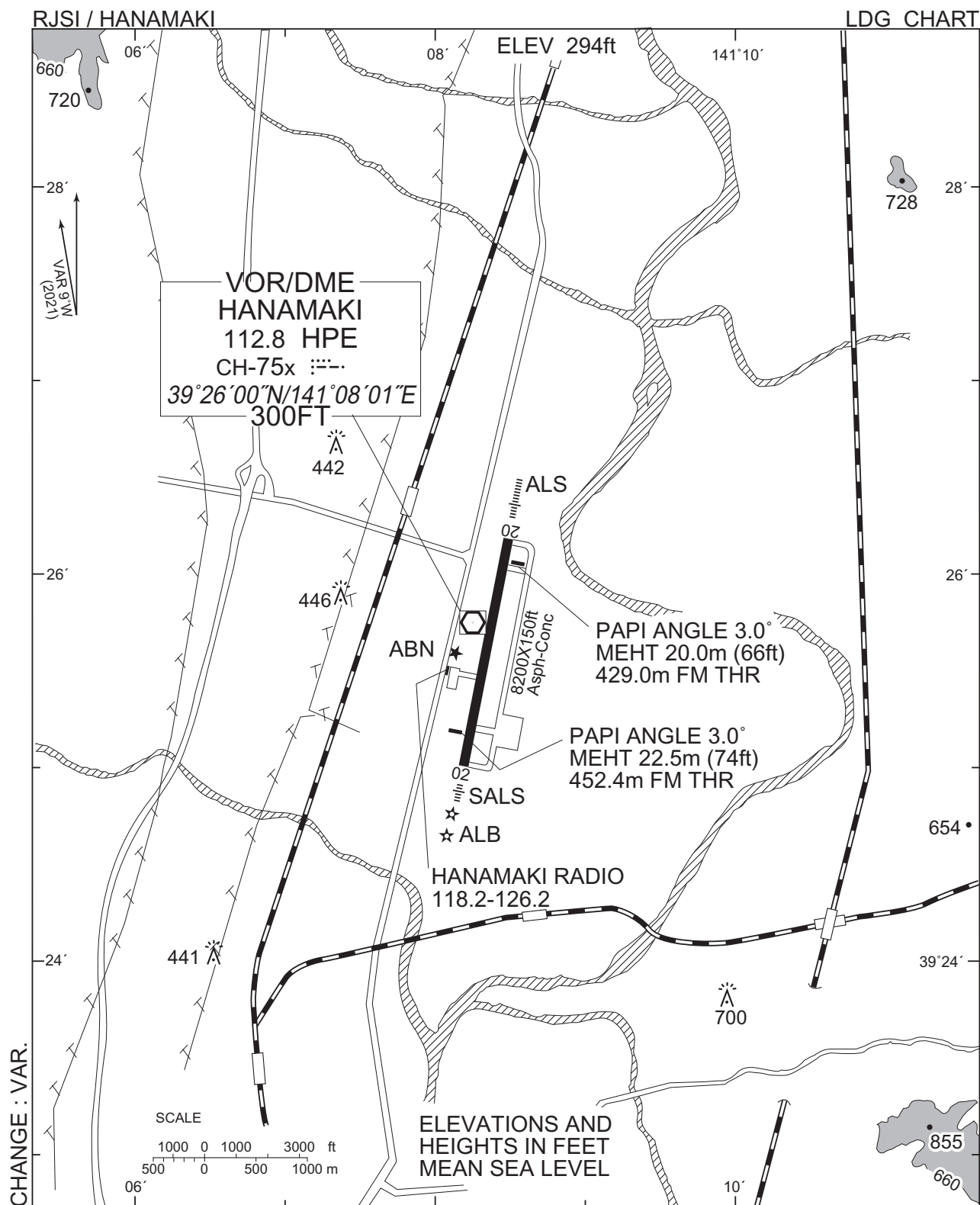
Visual REP



※図中に標高を示す数字がある場合、単位はメートル(m)である。The unit of measurement used to express elevation is meter(m).

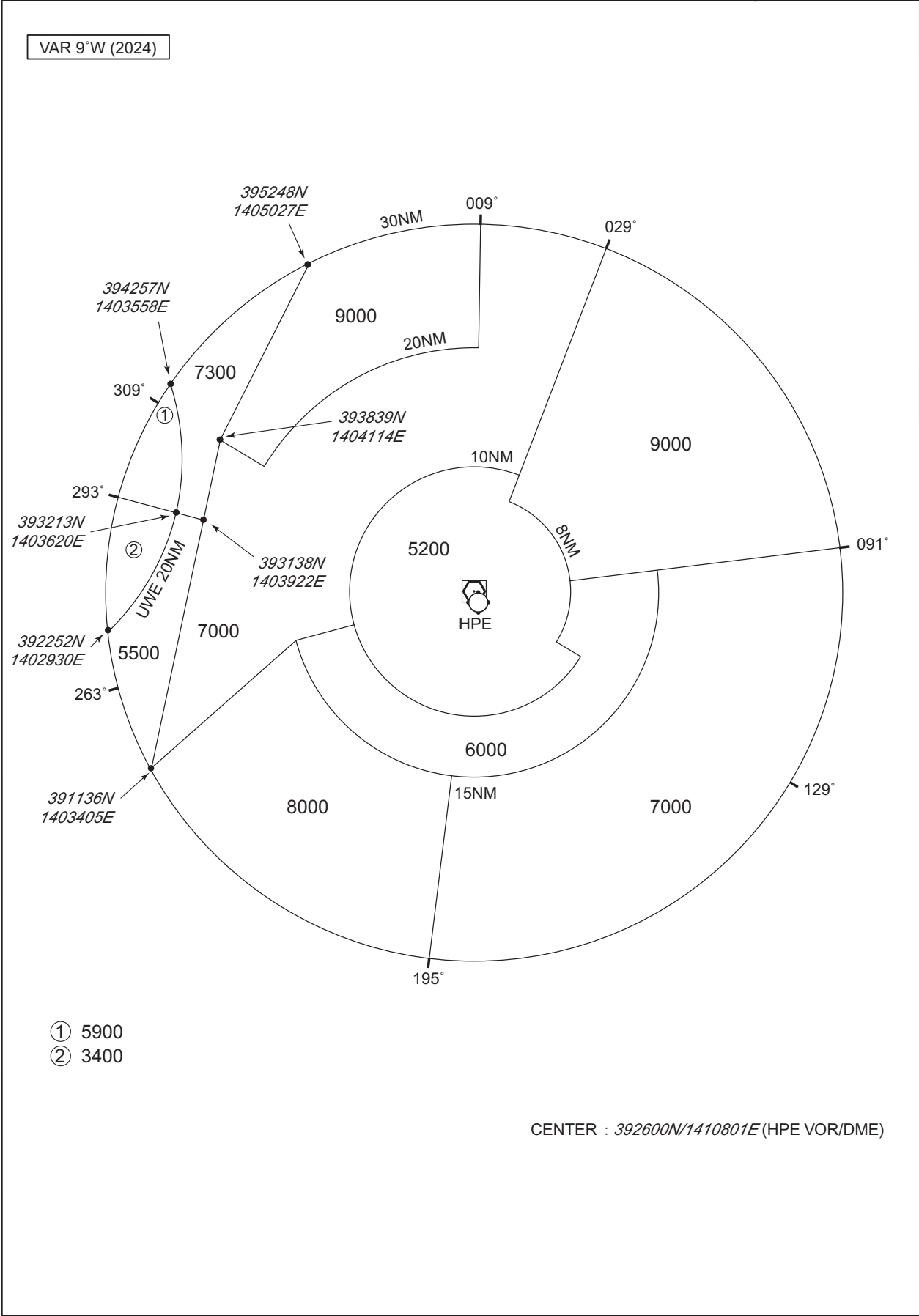
CHANGE : Map updated. BRG/DIST from ARP. Taseko established. Tsuchisawa abolished.

Call sign	BRG / DIST from ARP	Remarks
盛岡 Morioka	360°T / 16.4NM	JR駅 JR Station
城山 Shiroyama	012°T / 8.4NM	城跡 The site of a castle
豊沢 Toyosawa	293°T / 8.1NM	豊沢ダム Dam
田瀬湖 Taseko	121°T / 10.0NM	田瀬ダム Dam
北上 Kitakami	184°T / 8.9NM	JR駅 JR Station
水沢 Mizusawa	178°T / 17.4NM	JR駅 JR Station



RJSI / HANAMAKI

Minimum Vectoring Altitude CHART



CHANGE : Update.